HITEC UNIVERSITY TAXILA

DATABASE SYSTEM

SEMESTER 05
BSCS "C"

PROJECT REPORT

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Report on Database Management System:

Pakistan Tourism Management System

Introduction:

The aim of this project was to design and implement a Database Management System (DBMS) for a tourism management system, named "Pakistan Tourism Management System" (PTMS). This system focuses on managing tourist data, bookings, accommodations, and guided tours, ensuring efficient organization and retrieval of information. The project incorporates database design, normalization, and the development of a user-friendly portal for CRUD operations.

Database Design:

1. Database Overview

The database includes 17 interconnected tables covering key components like tourists, bookings, packages, accommodations, and more. This structure allows robust data management, query capabilities, and scalability for future additions.

2. Tables and Relationships

Key tables include:

Tourists: Stores tourist details.

TouristPackages: Defines available packages and destinations.

Bookings: Captures bookings made by tourists.

Accommodations, Guides, Transportation: Handle related services.

Feedback and Payments: Store tourist reviews and financial transactions.

And 9 more…

Relationships:

One-to-Many: A tourist can make multiple bookings.

Many-to-Many: Destinations can have multiple guides and vice versa, managed via a junction table.

3. Normalization Process

The database was normalized to 3NF:

1NF: Eliminated multi-valued attributes (e.g., splitting ContactInfo in the Tourists table).

2NF: Removed partial dependencies (e.g., separating TotalAmount into a new table in Bookings).

3NF: Eliminated transitive dependencies (e.g., moving Event Costs to a separate table).

Refer to the provided SQL scripts for normalized table creation and data insertion.

4. ER Diagram

The ER diagram illustrates the relationships and constraints between entities like tourists, packages, and bookings.

Portal Development:

1. Technology Used

The portal was developed using Python and the Tkinter framework for a Windows-based GUI. The backend is connected to the SQL database.

2. User Interface

The portal offers:

Data Display: Listbox for browsing records.

Form-Based Inputs: Adding and updating records.

Navigation: Buttons for key actions like adding, deleting, and viewing details.

3. Features

Aesthetic design with modern fonts and color schemes.

Simulated records for testing functionality.

Integration with the database for dynamic operations.

CRUD Operations:

1. Create

New records can be added via a user-friendly form. Validation ensures data integrity (e.g., mandatory fields for name and contact).

2. Read

The portal retrieves and displays data using SQL SELECT queries with joins. A unified view combines data across tables for a holistic summary.

3. Update

Existing records can be edited using the update functionality, ensuring data consistency.

4. Delete

Records can be safely removed with proper prompts to prevent accidental deletions.

Error Handling:

The application validates inputs, manages database errors (e.g., foreign key violations), and provides meaningful messages to users.

TASK 1:

- ## SQL QUERY BEFORE NORMALIZATION:

```sql
create database pakistan_tourist_management;

CREATE TABLE Tourists (
    TouristID INT PRIMARY KEY,
    Name VARCHAR(100),
    ContactInfo VARCHAR(255),
    Nationality VARCHAR(50)
);

CREATE TABLE TouristPackages (
    PackageID INT PRIMARY KEY,
    PackageName VARCHAR(100),
    DestinationDetails VARCHAR(255),
    Price DECIMAL(10, 2)
);

CREATE TABLE Bookings (
    BookingID INT PRIMARY KEY,
    TouristID INT,
    PackageID INT,
    BookingDate DATE,
    TotalAmount DECIMAL(10, 2)
```

```sql
);

CREATE TABLE Destinations (
    DestinationID INT PRIMARY KEY,
    DestinationName VARCHAR(100),
    Region VARCHAR(50),
    TouristSpots VARCHAR(255)
);

CREATE TABLE Guides (
    GuideID INT PRIMARY KEY,
    Name VARCHAR(100),
    Experience VARCHAR(50),
    Language VARCHAR(50),
    DestinationName VARCHAR(100)
);

CREATE TABLE Accommodations (
    AccommodationID INT PRIMARY KEY,
    Name VARCHAR(100),
    Location VARCHAR(100),
    CostPerNight DECIMAL(10, 2)
);

CREATE TABLE TouristFeedback (
    FeedbackID INT PRIMARY KEY,
    TouristID INT,
    BookingID INT,
    FeedbackText TEXT
);

CREATE TABLE Transportation (
    TransportID INT PRIMARY KEY,
    Type VARCHAR(50),
    CompanyName VARCHAR(100),
    Cost DECIMAL(10, 2)
);

CREATE TABLE TouristPayments (
    PaymentID INT PRIMARY KEY,
    TouristID INT,
    PaymentDate DATE,
    Amount DECIMAL(10, 2)
);

CREATE TABLE GuidedTours (
    TourID INT PRIMARY KEY,
    GuideID INT,
    TouristID INT,
    Date DATE
);

CREATE TABLE CulturalEvents (
    EventID INT PRIMARY KEY,
    EventName VARCHAR(100),
    Location VARCHAR(100),
    Cost DECIMAL(10, 2)
);
```

```sql
CREATE TABLE EmergencyContacts (
    ContactID INT PRIMARY KEY,
    DestinationID INT,
    ContactName VARCHAR(100),
    Phone VARCHAR(15)
);

CREATE TABLE TouristGroups (
    GroupID INT PRIMARY KEY,
    TouristID INT,
    GroupName VARCHAR(100)
);

CREATE TABLE RoomBookings (
    RoomBookingID INT PRIMARY KEY,
    AccommodationID INT,
    TouristID INT,
    StartDate DATE,
    EndDate DATE
);


INSERT INTO Tourists (TouristID, Name, ContactInfo, Nationality)
VALUES
(1, 'Ali Ahmed', '03211234567, ali@x.com', 'Pakistani'),
(2, 'Sarah Khan', '03121234567', 'Pakistani');

INSERT INTO TouristPackages (PackageID, PackageName, DestinationDetails, Price)
VALUES
(1, 'Northern Escape', 'Hunza, Skardu', 15000),
(2, 'Southern Safari', 'Karachi, Gwadar', 12000);

INSERT INTO Bookings (BookingID, TouristID, PackageID, BookingDate, TotalAmount)
VALUES
(1, 1, 1, '2024-01-01', 15000),
(2, 2, 2, '2024-01-10', 12000);

INSERT INTO Destinations (DestinationID, DestinationName, Region, TouristSpots)
VALUES
(1, 'Hunza', 'North', 'Attabad Lake, Karimabad'),
(2, 'Skardu', 'North', 'Shangrila, Deosai');

INSERT INTO Guides (GuideID, Name, Experience, Language, DestinationName)
VALUES
(1, 'Imran Khan', '5 years', 'Urdu', 'Hunza'),
(2, 'Zain Ali', '3 years', 'English', 'Skardu');

INSERT INTO Accommodations (AccommodationID, Name, Location, CostPerNight)
VALUES
(1, 'Hotel Hunza View', 'Hunza', 5000),
(2, 'Skardu Inn', 'Skardu', 4000);

INSERT INTO TouristFeedback (FeedbackID, TouristID, BookingID, FeedbackText)
VALUES
(1, 1, 1, 'Amazing experience with the Northern Escape package!'),
(2, 2, 2, 'Loved the Southern Safari!');
```

```sql
INSERT INTO Transportation (TransportID, Type, CompanyName, Cost)
VALUES
(1, 'Bus', 'Northern Tours', 2000),
(2, 'Car', 'Southern Wheels', 5000);

INSERT INTO TouristPayments (PaymentID, TouristID, PaymentDate, Amount)
VALUES
(1, 1, '2024-01-01', 15000),
(2, 2, '2024-01-10', 12000);

INSERT INTO GuidedTours (TourID, GuideID, TouristID, Date)
VALUES
(1, 1, 1, '2024-01-02'),
(2, 2, 2, '2024-01-11');

INSERT INTO CulturalEvents (EventID, EventName, Location, Cost)
VALUES
(1, 'Music Fest', 'Lahore', 5000),
(2, 'Food Expo', 'Karachi', 3000);

INSERT INTO EmergencyContacts (ContactID, DestinationID, ContactName, Phone)
VALUES
(1, 1, 'Local Police', '1234567890'),
(2, 2, 'Rescue Team', '0987654321');

INSERT INTO TouristGroups (GroupID, TouristID, GroupName)
VALUES
(1, 1, 'Family Vacation'),
(2, 2, 'Friends Trip');

INSERT INTO RoomBookings (RoomBookingID, AccommodationID, TouristID, StartDate, EndDate)
VALUES
(1, 1, 1, '2024-01-01', '2024-01-05'),
(2, 2, 2, '2024-01-10', '2024-01-15');
```

- <u>TABLE:</u>

1. Tourists Table

TouristID	Name	ContactInfo	Nationality
1	Ali Ahmed	03211234567, ali@x.com	Pakistani
2	Sarah Khan	03121234567	Pakistani

2. Tourist Packages Table

PackageID	PackageName	DestinationDetails	Price
1	Northern Escape	Hunza, Skardu	15000
2	Southern Safari	Karachi, Gwadar	12000

3. Bookings Table

BookingID	TouristID	PackageID	BookingDate	TotalAmount
1	1	1	2024-01-01	15000
2	2	2	2024-01-10	12000

4. Destinations Table

DestinationID	DestinationName	Region	TouristSpots
1	Hunza	North	Attabad Lake, Karimabad
2	Skardu	North	Shangrila, Deosai

5. Guides Table

GuideID	Name	Experience	Language	DestinationName
1	Imran Khan	5 years	Urdu	Hunza
2	Zain Ali	3 years	English	Skardu

6. Accommodations Table

AccommodationID	Name	Location	CostPerNight
1	Serena Hotel	Hunza	10000
2	Shangrila Resort	Skardu	15000

7. Tourist Feedback Table

FeedbackID	TouristID	BookingID	FeedbackText
1	1	1	Amazing experience!
2	2	2	Very enjoyable trip

8. Transportation Table

TransportID	Type	CompanyName	Cost
1	Bus	Daewoo	2000
2	Plane	PIA	10000

9. TouristPayments Table

PaymentID	TouristID	PaymentDate	Amount
1	1	2024-01-01	15000
2	2	2024-01-10	12000

10. GuidedTours Table

TourID	GuideID	TouristID	Date
1	1	1	2024-01-02
2	2	2	2024-01-12

11. CulturalEvents Table

EventID	EventName	Location	Cost
1	Music Fest	Lahore	5000
2	Food Expo	Karachi	3000

12. EmergencyContacts Table

ContactID	DestinationID	ContactName	Phone
1	1	Local Police	1234567890
2	2	Rescue Team	0987654321

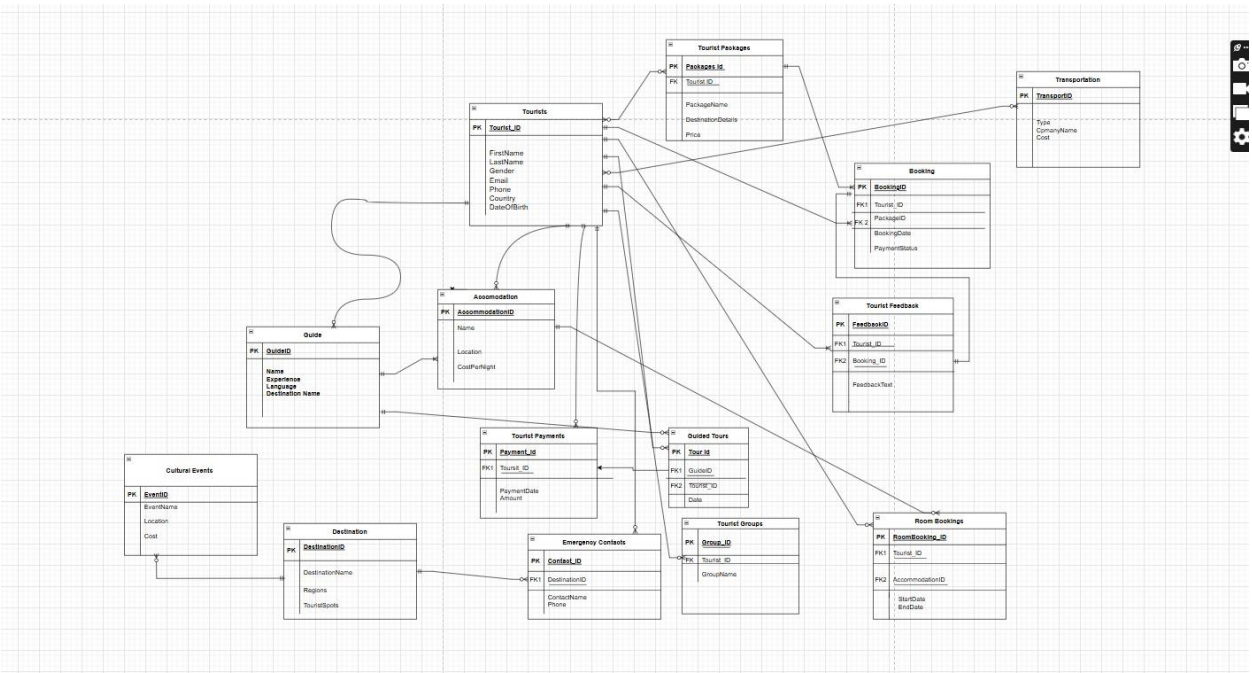
13. TouristGroups Table

GroupID	TouristID	GroupName
1	1	Group A
2	2	Group B

14. RoomBookings Table

RoomBookingID	AccommodationID	TouristID	StartDate	EndDate
1	1	1	2024-01-01	2024-01-05
2	2	2	2024-01-10	2024-01-15

- **<u>ERD FOR THE ABOVE TABLES:</u>**



- **<u>NORMALIZED FORM OF THE ABOVE ERD</u>**

Normalization to 1NF (Eliminate Multi-Valued Attributes):

1. Tourists Table

Issue: Multi-valued `ContactInfo`.

1NF Solution: Split multi-valued `ContactInfo` into separate rows.

TouristID	Name	ContactInfo	Nationality
1	Ali Ahmed	03211234567, ali@x.com	Pakistani
2	Sarah Khan	03121234567	Pakistani

Updated Table:

TouristID	Name	ContactInfo	Nationality
1	Ali Ahmed	03211234567	Pakistani
1	Ali Ahmed	ali@x.com	Pakistani
2	Sarah Khan	03121234567	Pakistani

2. TouristPackages Table

Issue: Non-atomic `DestinationDetails`.

1NF Solution: Break `DestinationDetails` into individual rows.

PackageID	PackageName	DestinationDetails	Price
1	Northern Escape	Hunza, Skardu	15000
2	Southern Safari	Karachi, Gwadar	12000

Updated Table:

PackageID	PackageName	DestinationDetail	Price
1	Northern Escape	Hunza	15000
1	Northern Escape	Skardu	15000
2	Southern Safari	Karachi	12000
2	Southern Safari	Gwadar	12000

3. Destinations Table

Issue: Multi-valued `TouristSpots`.

1NF Solution: Split `TouristSpots` into separate rows.

DestinationID	DestinationName	Region	TouristSpots
1	Hunza	North	Attabad Lake, Karimabad
2	Skardu	North	Shangrila, Deosai

Updated Table:

DestinationID	DestinationName	Region	TouristSpots
1	Hunza	North	Attabad Lake
1	Hunza	North	Karimabad
2	Skardu	North	Shangrila
2	Skardu	North	Deosai

ONLY THESE THREE TABLES HAVE MULTIVALUED ATTRIBUTE , NOW QUERY IS NORMALIZED TO 1NF.

Normalization to 2NF (Remove Partial Dependencies):

3. Bookings Table

Issue: `TotalAmount` depends only on `PackageID` (not the full composite key `BookingID`).

2NF Solution: Separate `TotalAmount` into a new table with `PackageID` as the key.

BookingID	TouristID	PackageID	BookingDate	TotalAmount
1	1	1	2024-01-01	15000
2	2	2	2024-01-10	12000

Updated Tables:

1.Bookings Table

BookingID	TouristID	PackageID	BookingDate
1	1	1	2024-01-01
2	2	2	2024-01-10

2. PackagePricing Table

PackageID	TotalAmount
1	15000
2	12000

5. Guides Table

Issue: `DestinationName` depends only on `GuideID` (not the full composite key).

2NF Solution: Split `DestinationName` into a new table.

GuideID	Name	Experience	Language	DestinationName
1	Imran Khan	5 years	Urdu	Hunza
2	Zain Ali	3 years	English	Skardu

Updated Tables:

1. Guides Table

GuideID	Name	Experience	Language
1	Imran Khan	5 years	Urdu
2	Zain Ali	3 years	English

2. GuideDestinations Table

DestinationID	GuideID	DestinationName
1	1	Lahore
2	2	Karachi

REST ALL ARE ALREADY CORRECT , MEANS NO PARTIAL DEPENDENCY

Normalization to 3NF (Remove Transitive Dependencies):

5. CulturalEvents Table

Issue: `Cost` depends on `Location` (transitive dependency).

3NF Solution: Move `Cost` to a separate table.

EventID	EventName	Location	Cost
1	Music Fest	Lahore	5000
2	Food Expo	Karachi	3000

Updated Tables:

1. CulturalEvents Table

EventID	EventName	Location
1	Music Fest	Lahore
2	Food Expo	Karachi

2. EventCosts Table

Location	Cost
Lahore	5000
Karachi	3000

6. EmergencyContacts Table

Issue: `ContactName` depends on `DestinationID` (transitive dependency).

3NF Solution: Split `ContactName` into a new table.

ContactID	DestinationID	ContactName	Phone
1	1	Local Police	1234567890
2	2	Rescue Team	0987654321

Updated Tables:

1. EmergencyContacts Table

ContactID	DestinationID	Phone
1	1	1234567890
2	2	0987654321

2. ContactNames Table

DestinationID	ContactName
1	Local Police
2	Rescue Team

- **<u>TABLES AFTER NORMALIZATION:</u>**

1.Tourists Table

TouristID	Name	ContactInfo	Nationality
1	Ali Ahmed	03211234567	Pakistani
1	Ali Ahmed	ali@x.com	Pakistani
2	Sarah Khan	03121234567	Pakistani

2.TouristPackages Table

PackageID	PackageName	DestinationDetail	Price
1	Northern Escape	Hunza	15000
1	Northern Escape	Skardu	15000
2	Southern Safari	Karachi	12000
2	Southern Safari	Gwadar	12000

3.Bookings Table

BookingID	TouristID	PackageID	BookingDate
1	1	1	2024-01-01
2	2	2	2024-01-10

4.PackagePricing Table

PackageID	TotalAmount
1	15000
2	12000

5.Destinations Table

DestinationID	DestinationName	Region	TouristSpots
1	Hunza	North	Attabad Lake
1	Hunza	North	Karimabad
2	Skardu	North	Shangrila
2	Skardu	North	Deosai

6.GUIDE TABLE:

GuideID	Name	Experience	Language
1	Imran Khan	5 years	Urdu
2	Zain Ali	3 years	English

7.GuideDestinations Table

DestinationID	GuideID	DestinationName
1	1	Lahore
2	2	Karachi

8.Accommodations Table

AccommodationID	Name	Location	CostPerNight
1	Serena Hotel	Hunza	10000
2	Shangrila Resort	Skardu	15000

9.Tourist Feedback Table

FeedbackID	TouristID	BookingID	FeedbackText
1	1	1	Amazing experience!
2	2	2	Very enjoyable trip

10.Transportation Table

TransportID	Type	CompanyName	Cost
1	Bus	Daewoo	2000
2	Plane	PIA	10000

11.TouristPayments Table

PaymentID	TouristID	PaymentDate	Amount
1	1	2024-01-01	15000
2	2	2024-01-10	12000

12.CulturalEvents Table

EventID	EventName	Location
1	Music Fest	Lahore
2	Food Expo	Karachi

13.EventCosts Table

Location	Cost
Lahore	5000
Karachi	3000

14.EmergencyContacts Table

ContactID	DestinationID	Phone
1	1	1234567890
2	2	0987654321

15.ContactNames Table

DestinationID	ContactName
1	Local Police
2	Rescue Team

16.TouristGroups Table

GroupID	TouristID	GroupName
1	1	Group A
2	2	Group B

17.RoomBookings Table

RoomBookingID	AccommodationID	TouristID	StartDate	EndDate
1	1	1	2024-01-01	2024-01-05
2	2	2	2024-01-10	2024-01-15

- **SQL FOR NORMALIZED DATABASE:**

```sql
-- Create Database
CREATE DATABASE PTMS1;
USE PTMS1;

-- Create tables
CREATE TABLE Tourists (
    TouristID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    ContactInfo VARCHAR(100) NOT NULL,
    Nationality VARCHAR(50) NOT NULL
);

CREATE TABLE TouristPackages (
    PackageID INT NOT NULL,
    PackageName VARCHAR(100) NOT NULL,
    Destination VARCHAR(100) NOT NULL,
    Price DECIMAL(10, 2) NOT NULL,
    PRIMARY KEY (PackageID, Destination)
);

CREATE TABLE Bookings (
    BookingID INT PRIMARY KEY,
    TouristID INT NOT NULL,
    PackageID INT NOT NULL,
    Destination VARCHAR(100) NOT NULL,
    BookingDate DATE NOT NULL,
    FOREIGN KEY (TouristID) REFERENCES Tourists(TouristID),
    FOREIGN KEY (PackageID, Destination) REFERENCES TouristPackages(PackageID,
Destination)
);

CREATE TABLE PackagePricing (
    PackageID INT NOT NULL,
    Destination VARCHAR(100) NOT NULL,
    TotalAmount DECIMAL(10, 2) NOT NULL,
    PRIMARY KEY (PackageID, Destination),
    FOREIGN KEY (PackageID, Destination) REFERENCES TouristPackages(PackageID,
Destination)
);

CREATE TABLE Destinations (
    DestinationID INT PRIMARY KEY,
    DestinationName VARCHAR(100) NOT NULL,
```

```sql
    Region VARCHAR(50) NOT NULL,
    TouristSpots VARCHAR(50) NOT NULL
);

CREATE TABLE Guides (
    GuideID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Experience VARCHAR(50) NOT NULL,
    Language VARCHAR(50) NOT NULL
);

CREATE TABLE GuideDestinations (
    DestinationID INT NOT NULL,
    GuideID INT NOT NULL,
    PRIMARY KEY (DestinationID, GuideID),
    FOREIGN KEY (DestinationID) REFERENCES Destinations(DestinationID),
    FOREIGN KEY (GuideID) REFERENCES Guides(GuideID)
);

CREATE TABLE Accommodations (
    AccommodationID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Location VARCHAR(100) NOT NULL,
    CostPerNight DECIMAL(10, 2) NOT NULL
);

CREATE TABLE TouristFeedback (
    FeedbackID INT PRIMARY KEY,
    TouristID INT NOT NULL,
    BookingID INT NOT NULL,
    FeedbackText TEXT NOT NULL,
    FOREIGN KEY (TouristID) REFERENCES Tourists(TouristID),
    FOREIGN KEY (BookingID) REFERENCES Bookings(BookingID)
);

CREATE TABLE Transportation (
    TransportID INT PRIMARY KEY,
    Type VARCHAR(50) NOT NULL,
    CompanyName VARCHAR(100) NOT NULL,
    Cost DECIMAL(10, 2) NOT NULL
);

CREATE TABLE TouristPayments (
    PaymentID INT PRIMARY KEY,
    TouristID INT NOT NULL,
    PaymentDate DATE NOT NULL,
    Amount DECIMAL(10, 2) NOT NULL,
    FOREIGN KEY (TouristID) REFERENCES Tourists(TouristID)
);

CREATE TABLE CulturalEvents (
    EventID INT PRIMARY KEY,
    EventName VARCHAR(100) NOT NULL,
    Location VARCHAR(100) NOT NULL
);

CREATE TABLE EventCosts (
    Location VARCHAR(100) PRIMARY KEY,
```

```sql
    Cost DECIMAL(10, 2) NOT NULL
);

CREATE TABLE EmergencyContacts (
    ContactID INT PRIMARY KEY,
    DestinationID INT NOT NULL,
    Phone VARCHAR(15) NOT NULL,
    FOREIGN KEY (DestinationID) REFERENCES Destinations(DestinationID)
);

CREATE TABLE ContactNames (
    DestinationID INT PRIMARY KEY,
    ContactName VARCHAR(100) NOT NULL,
    FOREIGN KEY (DestinationID) REFERENCES Destinations(DestinationID)
);

CREATE TABLE TouristGroups (
    GroupID INT PRIMARY KEY,
    TouristID INT NOT NULL,
    GroupName VARCHAR(100) NOT NULL,
    FOREIGN KEY (TouristID) REFERENCES Tourists(TouristID)
);

CREATE TABLE RoomBookings (
    RoomBookingID INT PRIMARY KEY,
    AccommodationID INT NOT NULL,
    TouristID INT NOT NULL,
    StartDate DATE NOT NULL,
    EndDate DATE NOT NULL,
    FOREIGN KEY (AccommodationID) REFERENCES Accommodations(AccommodationID),
    FOREIGN KEY (TouristID) REFERENCES Tourists(TouristID)
);

-- Insert data into Tourists
INSERT INTO Tourists (TouristID, Name, ContactInfo, Nationality) VALUES
(1, 'Ali Khan', 'ali.khan@example.com', 'Pakistani'),
(2, 'John Smith', 'john.smith@example.com', 'British'),
(3, 'Maria Lopez', 'maria.lopez@example.com', 'Spanish');

-- Insert data into TouristPackages
INSERT INTO TouristPackages (PackageID, PackageName, Destination, Price) VALUES
(1, 'Northern Adventure', 'Hunza', 20000.00),
(2, 'Cultural Tour', 'Lahore', 15000.00),
(3, 'Beach Vacation', 'Karachi', 18000.00);

-- Insert data into Bookings
INSERT INTO Bookings (BookingID, TouristID, PackageID, Destination, BookingDate) VALUES
(1, 1, 1, 'Hunza', '2025-01-01'),
(2, 2, 2, 'Lahore', '2025-01-05'),
(3, 3, 3, 'Karachi', '2025-01-10');

-- Insert data into PackagePricing
INSERT INTO PackagePricing (PackageID, Destination, TotalAmount) VALUES
(1, 'Hunza', 25000.00),
(2, 'Lahore', 17000.00),
(3, 'Karachi', 20000.00);

-- Insert data into Destinations
```

```sql
INSERT INTO Destinations (DestinationID, DestinationName, Region, TouristSpots) VALUES
(1, 'Hunza', 'Northern Pakistan', 'Mountains'),
(2, 'Lahore', 'Punjab', 'Historical Sites'),
(3, 'Karachi', 'Sindh', 'Beaches');

-- Insert data into Guides
INSERT INTO Guides (GuideID, Name, Experience, Language) VALUES
(1, 'Ahmed Ali', '5 Years', 'Urdu'),
(2, 'James Brown', '3 Years', 'English'),
(3, 'Sofia Garcia', '7 Years', 'Spanish');

-- Insert data into GuideDestinations
INSERT INTO GuideDestinations (DestinationID, GuideID) VALUES
(1, 1),
(2, 2),
(3, 3);

-- Insert data into Accommodations
INSERT INTO Accommodations (AccommodationID, Name, Location, CostPerNight) VALUES
(1, 'Hunza Hotel', 'Hunza', 5000.00),
(2, 'Lahore Inn', 'Lahore', 3000.00),
(3, 'Beach Resort', 'Karachi', 4000.00);

-- Insert data into TouristFeedback
INSERT INTO TouristFeedback (FeedbackID, TouristID, BookingID, FeedbackText) VALUES
(1, 1, 1, 'Amazing experience in Hunza!'),
(2, 2, 2, 'Loved the cultural tour in Lahore.'),
(3, 3, 3, 'Beach vacation was relaxing.');

-- Insert data into Transportation
INSERT INTO Transportation (TransportID, Type, CompanyName, Cost) VALUES
(1, 'Bus', 'Northern Travels', 1500.00),
(2, 'Train', 'Pakistan Railways', 1000.00),
(3, 'Plane', 'PIA', 8000.00);

-- Insert data into TouristPayments
INSERT INTO TouristPayments (PaymentID, TouristID, PaymentDate, Amount) VALUES
(1, 1, '2025-01-02', 25000.00),
(2, 2, '2025-01-06', 17000.00),
(3, 3, '2025-01-11', 20000.00);

-- Insert data into CulturalEvents
INSERT INTO CulturalEvents (EventID, EventName, Location) VALUES
(1, 'Music Festival', 'Lahore'),
(2, 'Art Exhibition', 'Karachi'),
(3, 'Food Carnival', 'Hunza');

-- Insert data into EventCosts
INSERT INTO EventCosts (Location, Cost) VALUES
('Lahore', 5000.00),
('Karachi', 4000.00),
('Hunza', 3000.00);

-- Insert data into EmergencyContacts
INSERT INTO EmergencyContacts (ContactID, DestinationID, Phone) VALUES
(1, 1, '0345-1234567'),
(2, 2, '0300-9876543'),
(3, 3, '0321-5678901');
```

```sql
-- Insert data into ContactNames
INSERT INTO ContactNames (DestinationID, ContactName) VALUES
(1, 'Zahid Ali'),
(2, 'Ayesha Khan'),
(3, 'Bilal Ahmed');

-- Insert data into TouristGroups
INSERT INTO TouristGroups (GroupID, TouristID, GroupName) VALUES
(1, 1, 'Northern Explorers'),
(2, 2, 'Cultural Enthusiasts'),
(3, 3, 'Beach Lovers');

-- Insert data into RoomBookings
INSERT INTO RoomBookings (RoomBookingID, AccommodationID, TouristID, StartDate, EndDate)
VALUES
(1, 1, 1, '2025-01-01', '2025-01-03'),
(2, 2, 2, '2025-01-05', '2025-01-07'),
(3, 3, 3, '2025-01-10', '2025-01-12');


SELECT * FROM Tourists;
SELECT * FROM TouristPackages;
SELECT * FROM Bookings;
SELECT * FROM PackagePricing;
SELECT * FROM Destinations;
SELECT * FROM Guides;
SELECT * FROM GuideDestinations;
SELECT * FROM Accommodations;
SELECT * FROM TouristFeedback;
SELECT * FROM Transportation;
SELECT * FROM TouristPayments;
SELECT * FROM CulturalEvents;
SELECT * FROM EventCosts;
SELECT * FROM EmergencyContacts;
SELECT * FROM ContactNames;
SELECT * FROM TouristGroups;
SELECT * FROM RoomBookings;


-- Create View
CREATE VIEW FullTourismView AS
SELECT
    t.TouristID,
    t.Name AS TouristName,
    t.ContactInfo,
    t.Nationality,
    tp.PackageName,
    tp.Destination,
    tp.Price AS PackagePrice,
    b.BookingDate,
    p.TotalAmount AS PackageTotalAmount,
    d.DestinationName,
    d.Region,
    d.TouristSpots,
    a.Name AS AccommodationName,
    a.Location AS AccommodationLocation,
    a.CostPerNight,
```

```sql
    g.Name AS GuideName,
    g.Experience AS GuideExperience,
    g.Language AS GuideLanguage,
    tf.FeedbackText AS TouristFeedback,
    tr.Type AS TransportType,
    tr.CompanyName AS TransportCompany,
    tr.Cost AS TransportCost,
    pmt.PaymentDate,
    pmt.Amount AS PaymentAmount,
    ce.EventName AS CulturalEventName,
    ec.Cost AS EventCost,
    em.Phone AS EmergencyContactPhone,
    cn.ContactName AS EmergencyContactName,
    tg.GroupName AS TouristGroupName,
    rb.StartDate AS RoomBookingStart,
    rb.EndDate AS RoomBookingEnd
FROM
    Tourists t
JOIN Bookings b ON t.TouristID = b.TouristID
JOIN TouristPackages tp ON b.PackageID = tp.PackageID AND b.Destination = tp.Destination
JOIN PackagePricing p ON tp.PackageID = p.PackageID AND tp.Destination = p.Destination
JOIN Destinations d ON tp.Destination = d.DestinationName
LEFT JOIN Accommodations a ON d.DestinationName = a.Location
LEFT JOIN GuideDestinations gd ON d.DestinationID = gd.DestinationID
LEFT JOIN Guides g ON gd.GuideID = g.GuideID
LEFT JOIN TouristFeedback tf ON b.BookingID = tf.BookingID
LEFT JOIN Transportation tr ON tr.TransportID = 1 -- Adjust transport logic as needed
LEFT JOIN TouristPayments pmt ON t.TouristID = pmt.TouristID
LEFT JOIN CulturalEvents ce ON ce.Location = d.DestinationName
LEFT JOIN EventCosts ec ON ce.Location = ec.Location
LEFT JOIN EmergencyContacts em ON d.DestinationID = em.DestinationID
LEFT JOIN ContactNames cn ON d.DestinationID = cn.DestinationID
LEFT JOIN TouristGroups tg ON t.TouristID = tg.TouristID
LEFT JOIN RoomBookings rb ON t.TouristID = rb.TouristID;

-- Query the view to verify data
SELECT * FROM FullTourismView;
```

OUTPUT:

Results | Messages

	TouristID	Name	ContactInfo	Nationality
1	1	Ali Khan	ali.khan@example.com	Pakistani
2	2	John Smith	john.smith@example.com	British
3	3	Maria Lopez	maria.lopez@example.com	Spanish

	PackageID	PackageName	Destination	Price
1	1	Northern Adventure	Hunza	20000.00
2	2	Cultural Tour	Lahore	15000.00
3	3	Beach Vacation	Karachi	18000.00

	BookingID	TouristID	PackageID	Destination	BookingDate
1	1	1	1	Hunza	2025-01-01
2	2	2	2	Lahore	2025-01-05
3	3	3	3	Karachi	2025-01-10

	PackageID	Destination	TotalAmount
1	1	Hunza	25000.00
2	2	Lahore	17000.00
3	3	Karachi	20000.00

	DestinationID	DestinationName	Region	TouristSpots
1	1	Hunza	Northern Pakistan	Mountains
2	2	Lahore	Punjab	Historical Sites
3	3	Karachi	Sindh	Beaches

	GuideID	Name	Experience	Language
1	1	Ahmed Ali	5 Years	Urdu
2	2	James Brown	3 Years	English
3	3	Sofia Garcia	7 Years	Spanish

	DestinationID	GuideID
1	1	1
2	2	2
3	3	3

	AccommodationID	Name	Location	CostPerNight

Results | Messages

	AccommodationID	Name	Location	CostPerNight
1	1	Hunza Hotel	Hunza	5000.00
2	2	Lahore Inn	Lahore	3000.00
3	3	Beach Res...	Karachi	4000.00

	FeedbackID	TouristID	BookingID	FeedbackText
1	1	1	1	Amazing experience in Hunza!
2	2	2	2	Loved the cultural tour in Lahore.
3	3	3	3	Beach vacation was relaxing.

	TransportID	Type	CompanyName	Cost
1	1	Bus	Northern Travels	1500.00
2	2	Train	Pakistan Railways	1000.00
3	3	Pla...	PIA	8000.00

	PaymentID	TouristID	PaymentDate	Amount
1	1	1	2025-01-02	25000.00
2	2	2	2025-01-06	17000.00
3	3	3	2025-01-11	20000.00

	EventID	EventName	Location
1	1	Music Festival	Lahore
2	2	Art Exhibition	Karachi
3	3	Food Carnival	Hunza

	Location	Cost
1	Hunza	3000.00
2	Karachi	4000.00
3	Lahore	5000.00

	ContactID	DestinationID	Phone
1	1	1	0345-1234567
2	2	2	0300-9876543
3	3	3	0321-5678901

	DestinationID	ContactName

	DestinationID	ContactName
1	1	Zahid Ali
2	2	Ayesha Khan
3	3	Bilal Ahmed

	GroupID	TouristID	GroupName
1	1	1	Northern Explorers
2	2	2	Cultural Enthusiasts
3	3	3	Beach Lovers

	RoomBookingID	AccommodationID	TouristID	StartDate	EndDate
1	1	1	1	2025-01-01	2025-01-03
2	2	2	2	2025-01-05	2025-01-07
3	3	3	3	2025-01-10	2025-01-12

```sql
LEFT JOIN ContactNames cn ON d.DestinationID = cn.DestinationID
LEFT JOIN TouristGroups tg ON t.TouristID = tg.TouristID
LEFT JOIN RoomBookings rb ON t.TouristID = rb.TouristID;

-- Query the view to verify data
SELECT * FROM FullTourismView;
```

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Results | Messages

	TouristID	TouristName	ContactInfo	Nationality	PackageName	Destination	PackagePrice	BookingDate	PackageTotalAmount	DestinationName	Region	TouristSpo
1	1	Ali Khan	ali.khan@example.com	Pakistani	Northern Adventure	Hunza	20000.00	2025-01-01	25000.00	Hunza	Northern Pakistan	Mountains
2	2	John Smith	john.smith@example....	British	Cultural Tour	Lahore	15000.00	2025-01-05	17000.00	Lahore	Punjab	Historical .
3	3	Maria Lopez	maria.lopez@exampl...	Spanish	Beach Vacation	Karachi	18000.00	2025-01-10	20000.00	Karachi	Sindh	Beaches

TASK 2 AND 3:

For backend we had used python:

PYTHON CODE:

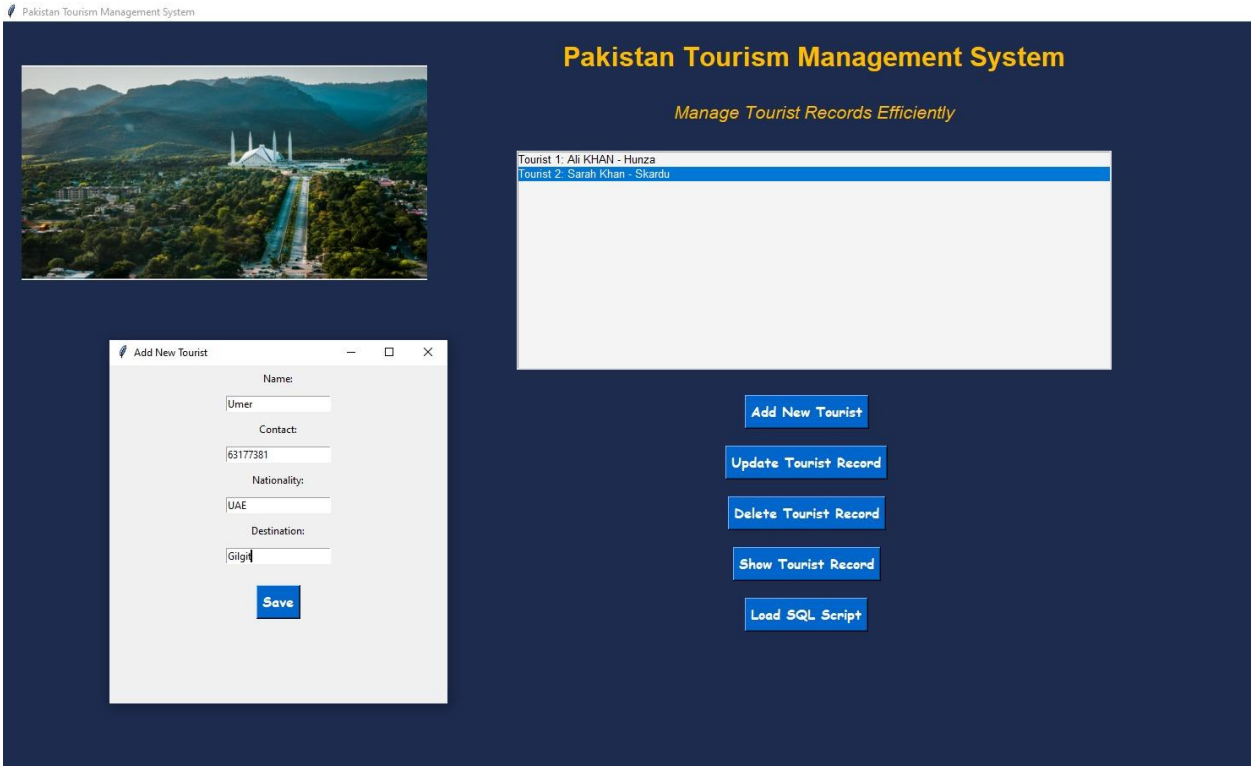
```python
import tkinter as tk
from tkinter import Label, Button, Listbox, messagebox, Entry, Toplevel, Text
import tkinter.ttk as ttk

# Tkinter GUI setup
root = tk.Tk()
root.title("Pakistan Tourism Management System")
root.geometry("1000x700")  # Set window size
root.configure(bg="#1D2B4F")  # Background color

# Fonts and Colors
heading_font = ('Helvetica', 24, 'bold')
subheading_font = ('Arial', 16, 'italic')
button_font = ('Comic Sans MS', 12, 'bold')
text_color = "#FFFFFF"
```

```
button_color = '#0066CC'
label_font = ('Courier', 12, 'bold')
entry_font = ('Verdana', 12)

# Heading
heading_label = Label(root, text="Pakistan Tourism Management System",
font=heading_font, bg="#1D2B4F", fg="#FFBC00")
heading_label.pack(pady=20)

# Subheading for instructions or actions
subheading_label = Label(root, text="Manage Tourist Records Efficiently",
font=subheading_font, bg="#1D2B4F", fg="#FFBC00")
subheading_label.pack(pady=10)

# Listbox to display records
listbox = Listbox(root, width=100, height=15, font=('Helvetica', 10),
bg="#F4F4F4", fg="black", selectmode="SINGLE")
listbox.pack(pady=20, padx=20)

# Scrollbar for Listbox
scrollbar = ttk.Scrollbar(root, orient="vertical", command=listbox.yview)
scrollbar.pack(side="right", fill="y")
listbox.config(yscrollcommand=scrollbar.set)

# Simulated tourist data
tourists_list = [
    {"id": 1, "name": "Ali Ahmed", "contact": "033455667", "nationality": "USA",
"package": "South Safaro", "destination": "Hunza", "price": 30000,
"booking_date": "11 JAN 2024", "guide": "Rehan", "experience": "4 years",
"hotel": "PC Hotel", "cost_per_night": 10000, "transportation": "Daewoo",
"emergency_contact": "2344585869", "group": "Group A"},
    {"id": 2, "name": "Sarah Khan", "contact": "033455668", "nationality":
"Pakistan", "package": "Mountain Trek", "destination": "Skardu", "price": 25000,
"booking_date": "15 FEB 2024", "guide": "Ahmed", "experience": "3 years",
"hotel": "Serena Hotel", "cost_per_night": 8000, "transportation": "Faisal
Movers", "emergency_contact": "2344585870", "group": "Group B"}
]

# Function to refresh the listbox with tourist data
def refresh_listbox():
    listbox.delete(0, 'end')
    for tourist in tourists_list:
        listbox.insert('end', f"Tourist {tourist['id']}: {tourist['name']} -
{tourist['destination']}")
```

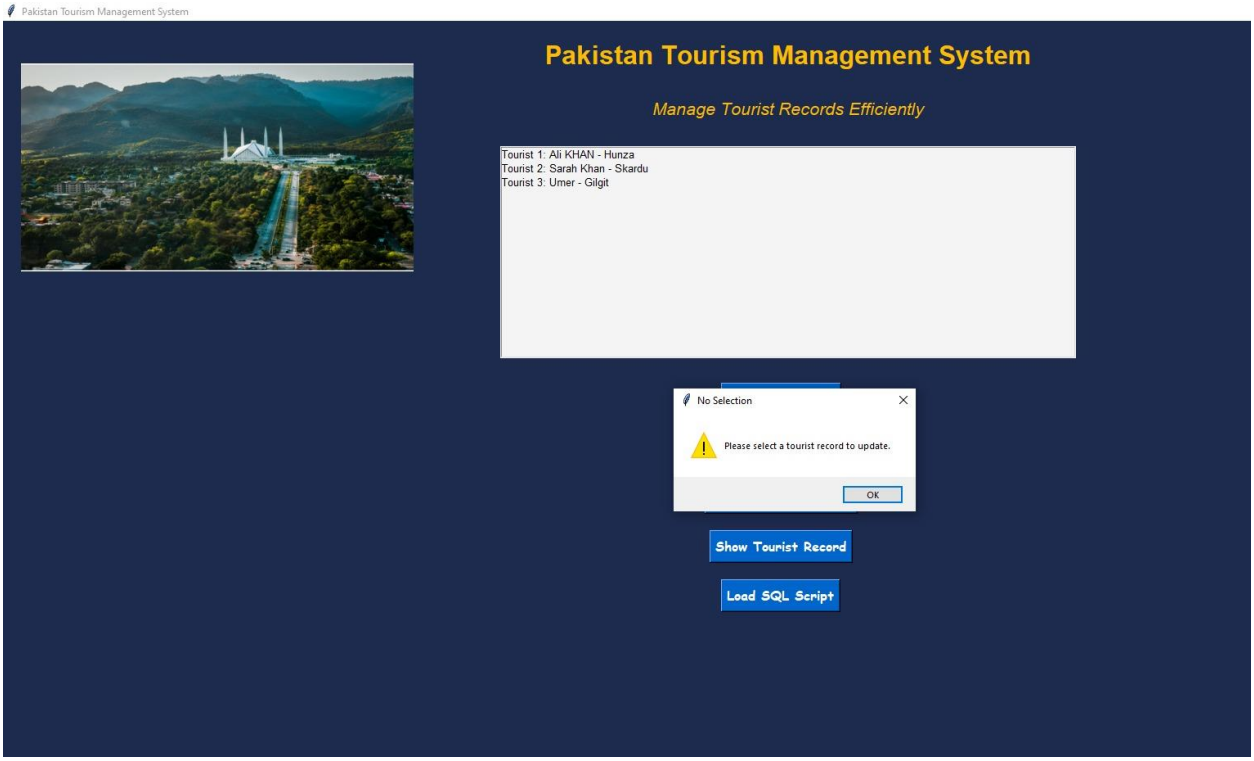
```
# Remaining functions: update_record, delete_record, show_record, load_sql_script
# (Omitted here for brevity, but include them as they are in your original code)

# Buttons
add_button = Button(root, text="Add New Tourist", command=None, font=button_font,
bg=button_color, fg=text_color)
add_button.pack(pady=10)

update_button = Button(root, text="Update Tourist Record", command=None,
font=button_font, bg=button_color, fg=text_color)
update_button.pack(pady=10)

delete_button = Button(root, text="Delete Tourist Record", command=None,
font=button_font, bg=button_color, fg=text_color)
delete_button.pack(pady=10)

show_button = Button(root, text="Show Tourist Record", command=None,
font=button_font, bg=button_color, fg=text_color)
show_button.pack(pady=10)

# SQL button to load and display SQL script
sql_button = Button(root, text="Load SQL Script", command=None, font=button_font,
bg=button_color, fg=text_color)
sql_button.pack(pady=10)

# Refresh listbox
refresh_listbox()

# Main loop
root.mainloop()
```

OUTPUT:

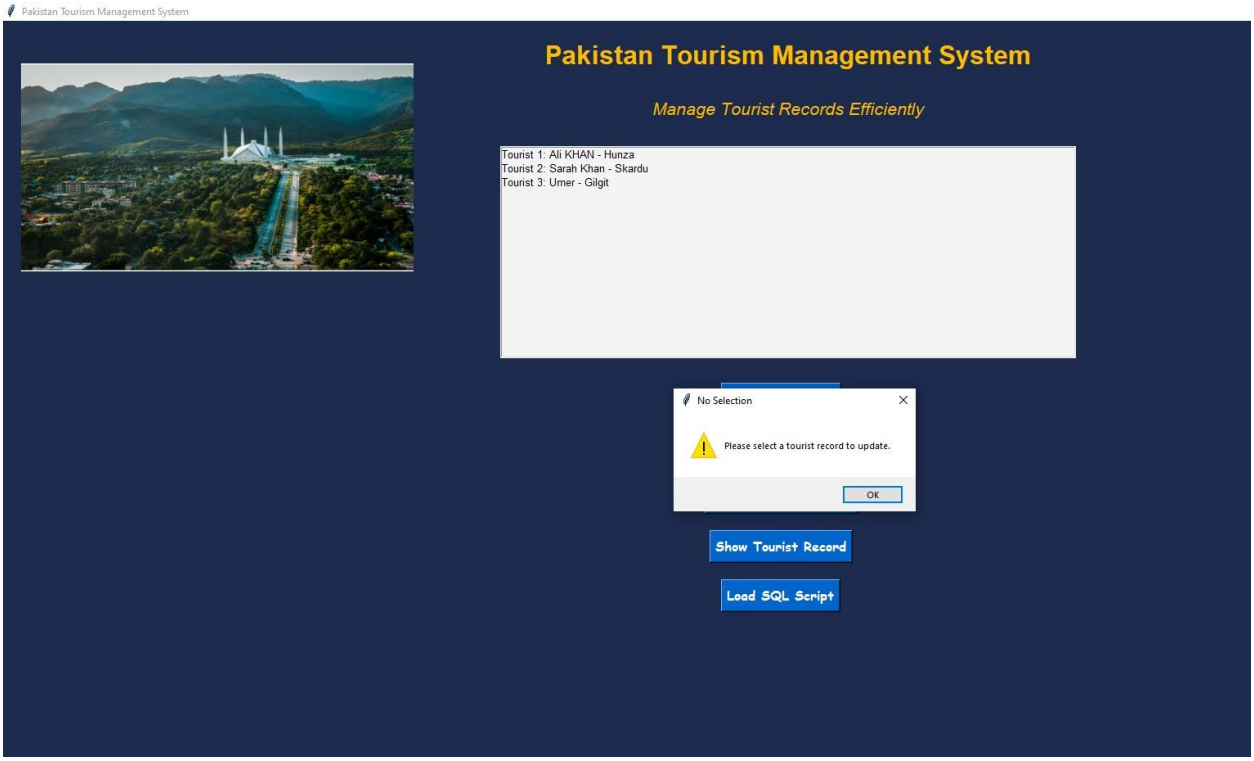
1.


Pakistan Tourism Management System
Pakistan Tourism Management System
Manage Tourist Records Efficiently
Tourist 1: Ali KHAN - Hunza
Tourist 2: Sarah Khan - Skardu
Add New Tourist
Name:
Umer
Contact:
63177381
Nationality:
UAE
Destination:
Gilgit
Save
Add New Tourist
Update Tourist Record
Delete Tourist Record
Show Tourist Record
Load SQL Script

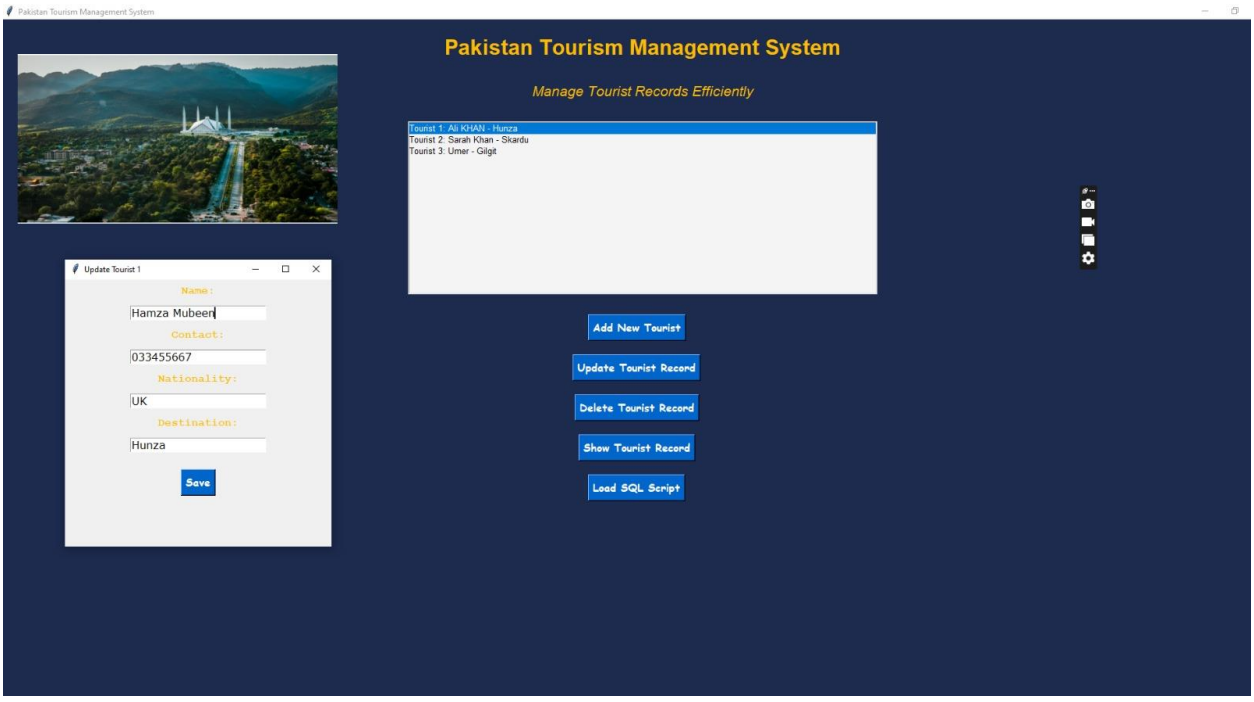
2.


Pakistan Tourism Management System
Pakistan Tourism Management System
Manage Tourist Records Efficiently
Tourist 1: Ali KHAN - Hunza
Tourist 2: Sarah Khan - Skardu
Tourist 3: Umer - Gilgit
No Selection
Please select a tourist record to update.
OK
Show Tourist Record
Load SQL Script

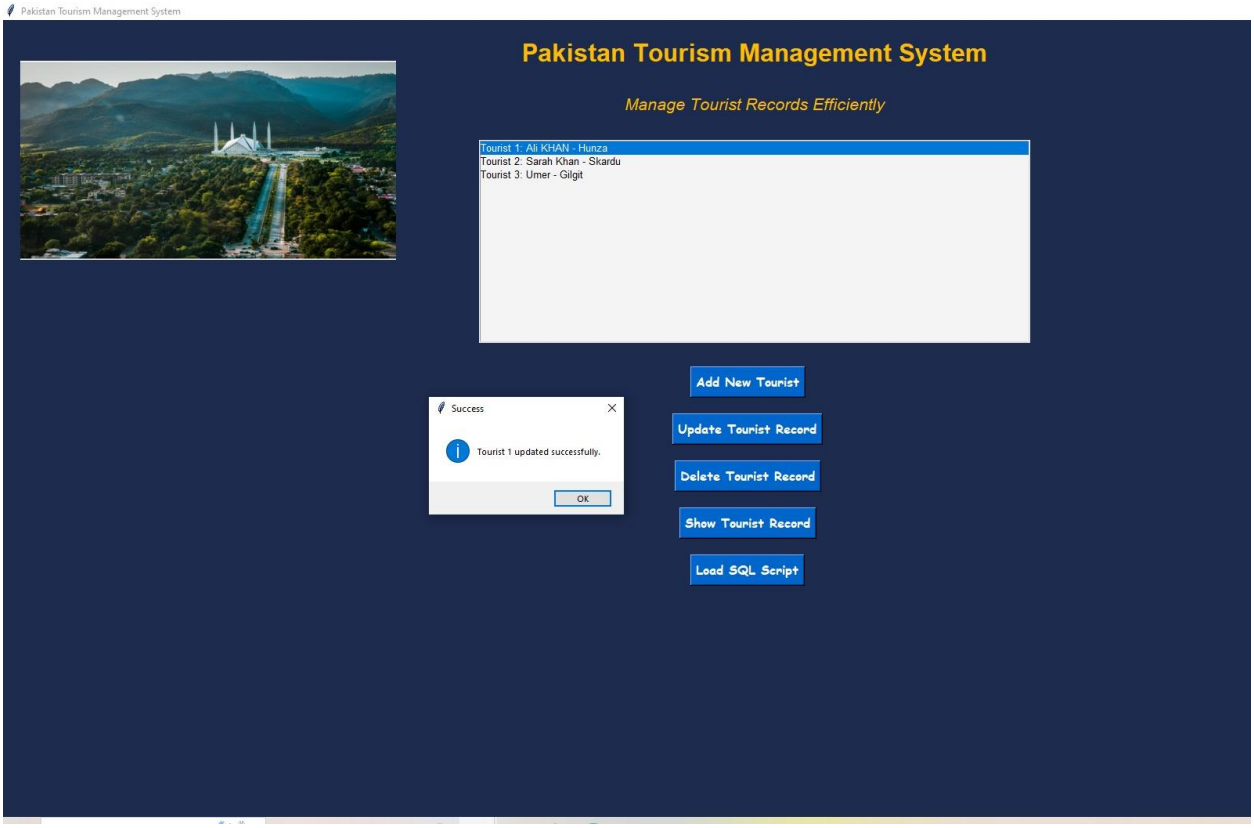
3.


Pakistan Tourism Management System
Pakistan Tourism Management System
Manage Tourist Records Efficiently
Tourist 1: Ali KHAN - Hunza
Tourist 2: Sarah Khan - Skardu
Tourist 3: Umer - Gilgit
No Selection
Please select a tourist record to update.
OK
Show Tourist Record
Load SQL Script

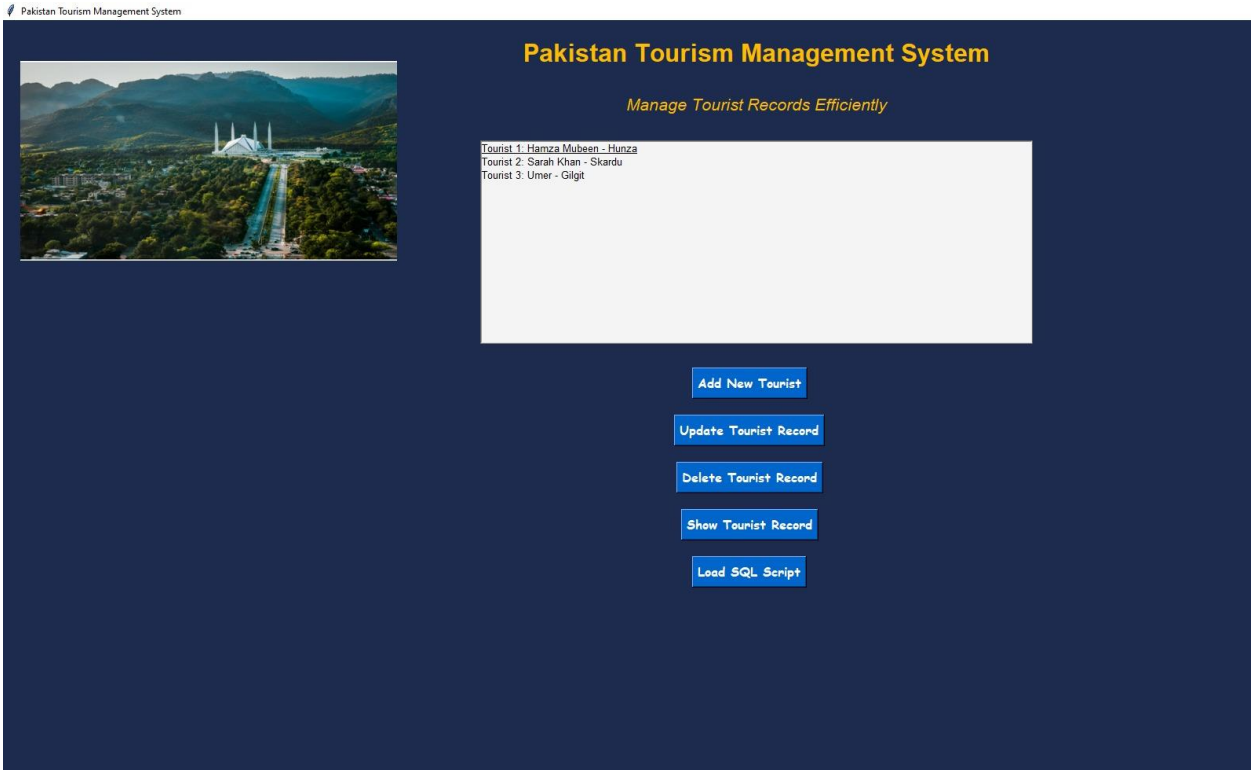
4.


Pakistan Tourism Management System
Pakistan Tourism Management System
Manage Tourist Records Efficiently
Tourist 1: Ali KHAN - Hunza
Tourist 2: Sarah Khan - Skardu
Tourist 3: Umer - Gilgit
Update Tourist 1
Name:
Hamza Mubeen
Contact:
033455667
Nationality:
UK
Destination:
Hunza
Save
Add New Tourist
Update Tourist Record
Delete Tourist Record
Show Tourist Record
Load SQL Script

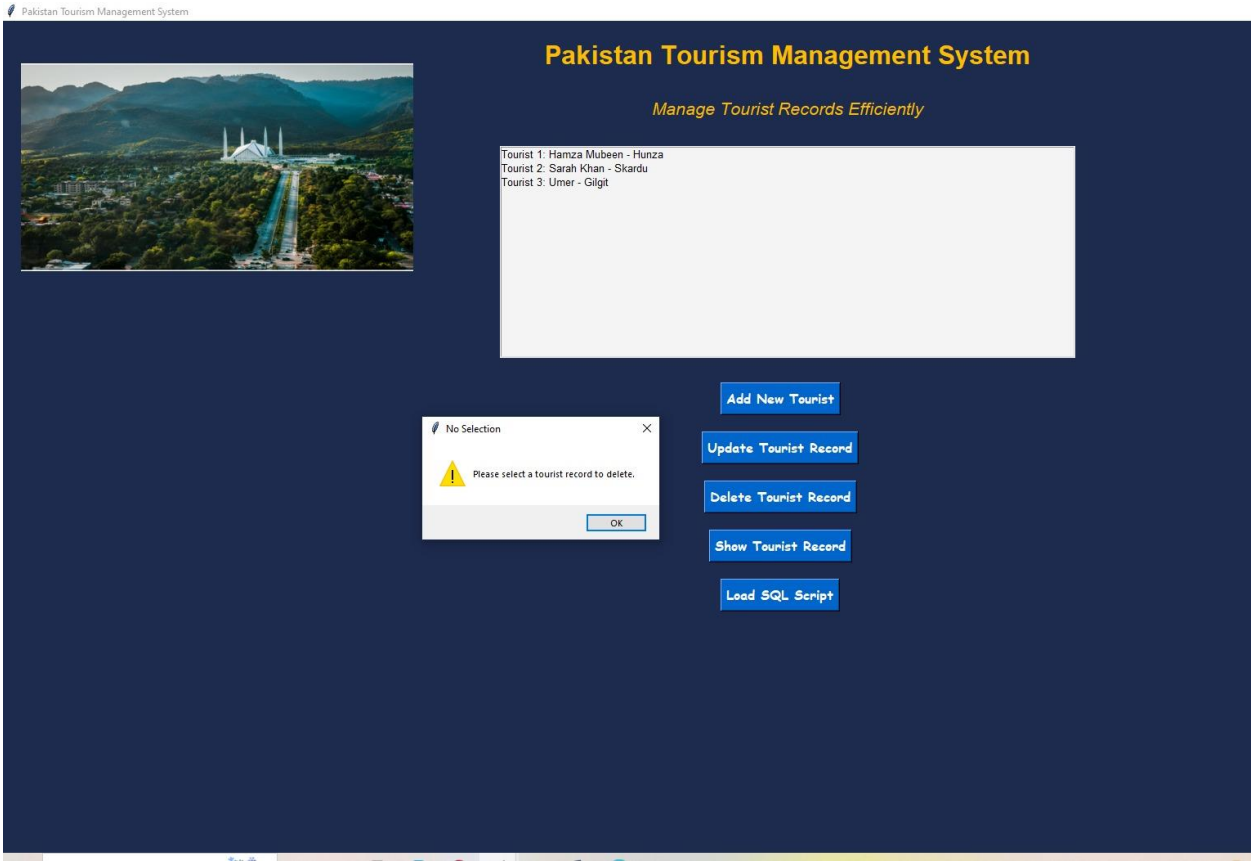
5.


Pakistan Tourism Management System
Pakistan Tourism Management System
Manage Tourist Records Efficiently
Tourist 1: Ali KHAN - Hunza
Tourist 2: Sarah Khan - Skardu
Tourist 3: Umer - Gilgit
Success
Tourist 1 updated successfully.
OK
Add New Tourist
Update Tourist Record
Delete Tourist Record
Show Tourist Record
Load SQL Script

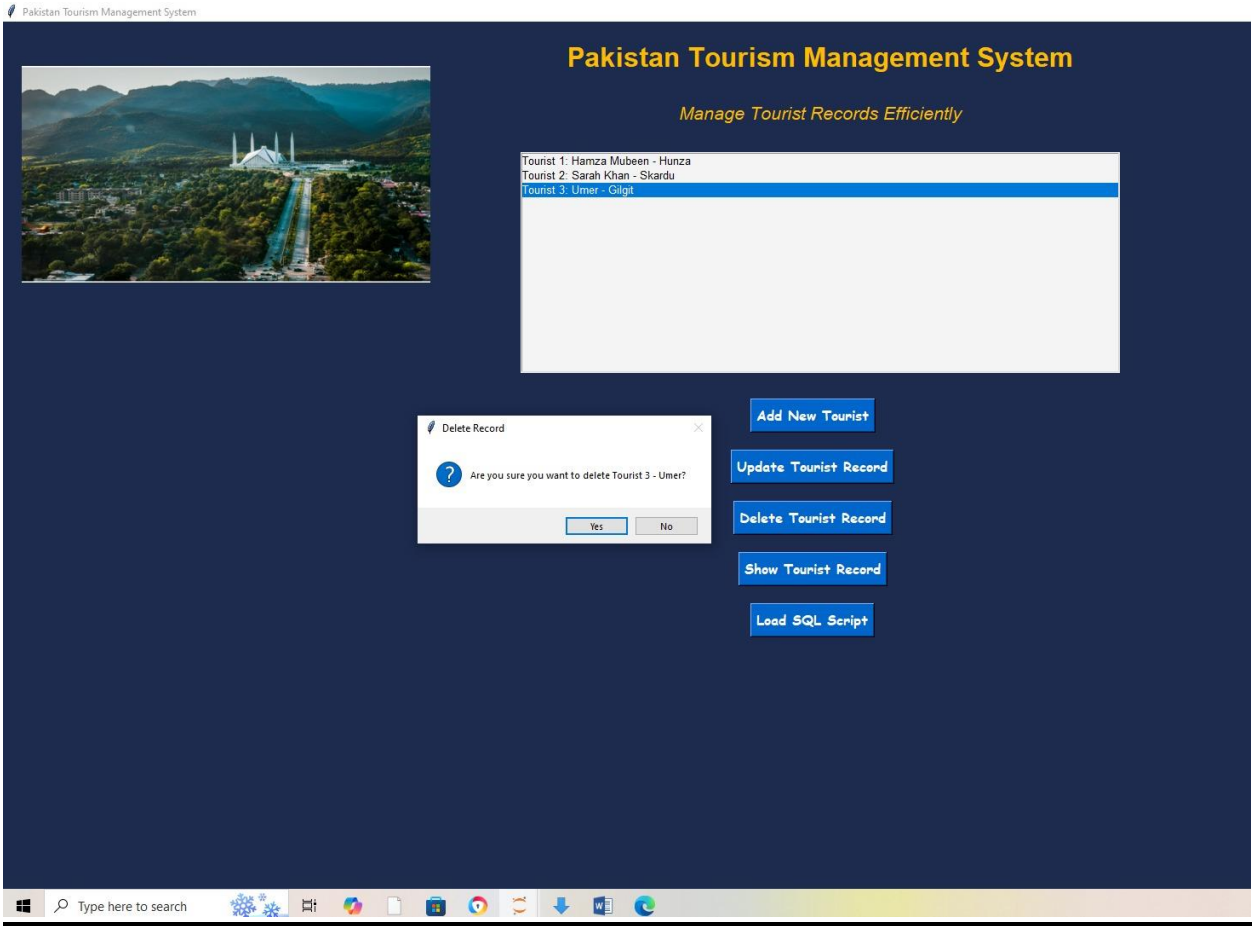
6.


Pakistan Tourism Management System
Pakistan Tourism Management System
Manage Tourist Records Efficiently
Tourist 1: Hamza Mubeen - Hunza
Tourist 2: Sarah Khan - Skardu
Tourist 3: Umer - Gilgit
Add New Tourist
Update Tourist Record
Delete Tourist Record
Show Tourist Record
Load SQL Script

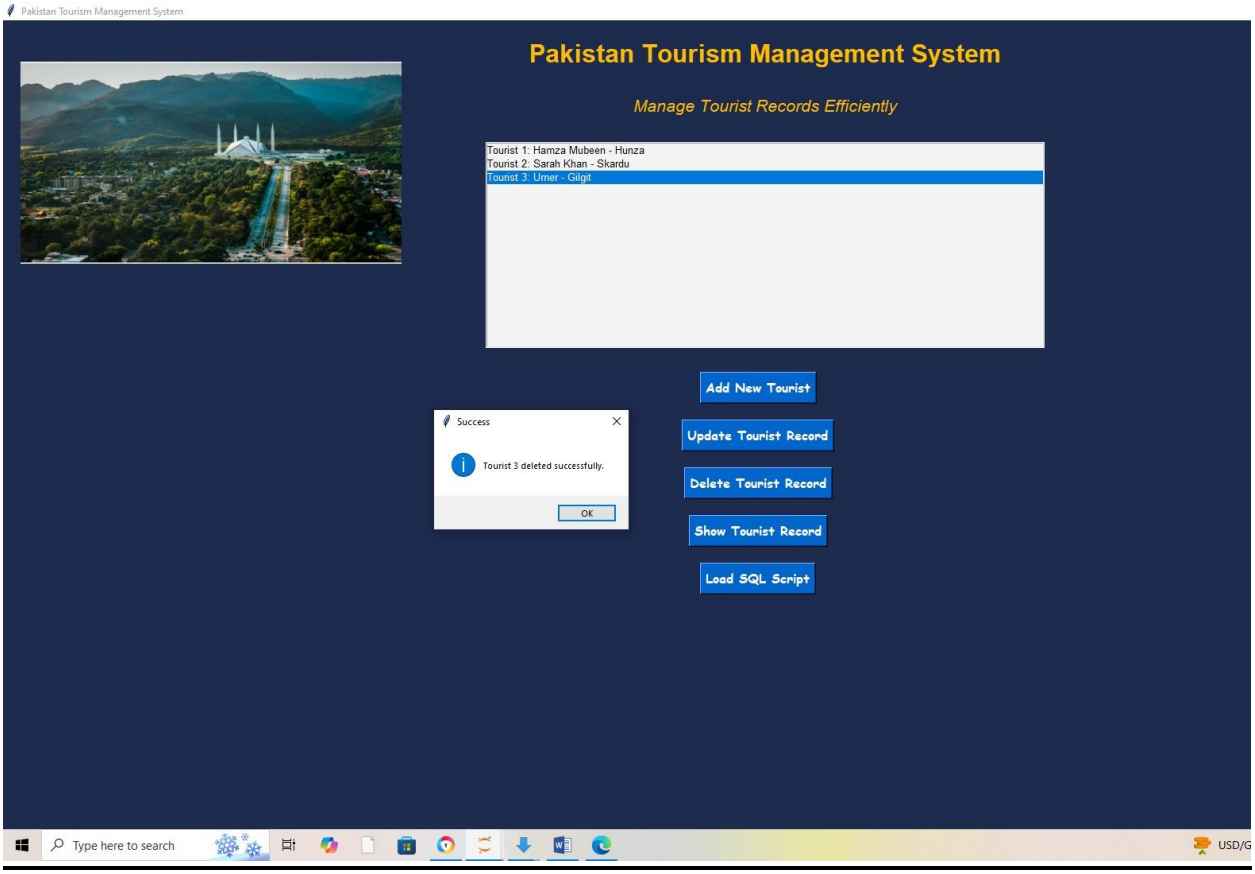
7.


Pakistan Tourism Management System
Pakistan Tourism Management System
Manage Tourist Records Efficiently
Tourist 1: Hamza Mubeen - Hunza
Tourist 2: Sarah Khan - Skardu
Tourist 3: Umer - Gilgit
No Selection
Please select a tourist record to delete.
OK
Add New Tourist
Update Tourist Record
Delete Tourist Record
Show Tourist Record
Load SQL Script

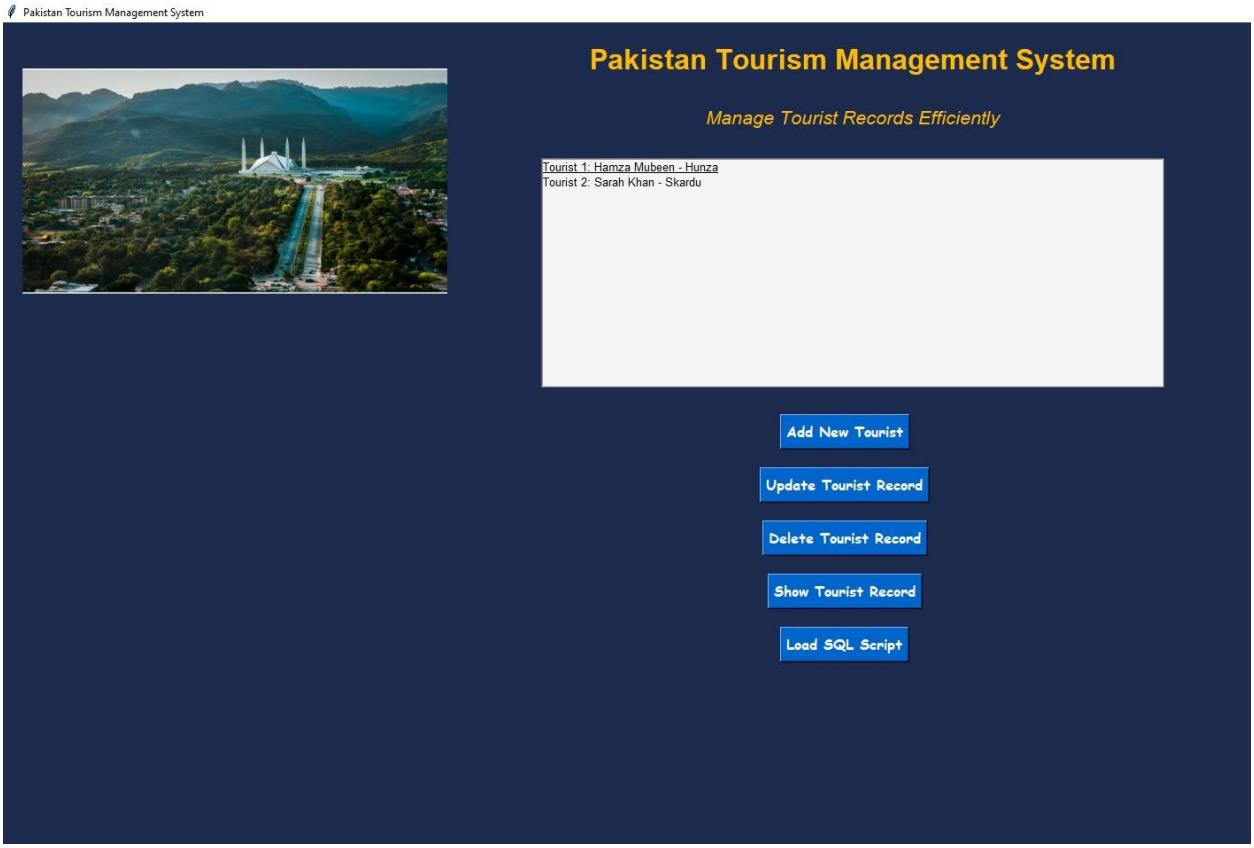
8.


Pakistan Tourism Management System
Pakistan Tourism Management System
Manage Tourist Records Efficiently
Tourist 1: Hamza Mubeen - Hunza
Tourist 2: Sarah Khan - Skardu
Tourist 3: Umer - Gilgit
Delete Record
Are you sure you want to delete Tourist 3 - Umer?
Yes
No
Add New Tourist
Update Tourist Record
Delete Tourist Record
Show Tourist Record
Load SQL Script
Type here to search

9.


Pakistan Tourism Management System
Pakistan Tourism Management System
Manage Tourist Records Efficiently
Tourist 1: Hamza Mubeen - Hunza
Tourist 2: Sarah Khan - Skardu
Tourist 3: Umer - Gilgit
Success
Tourist 3 deleted successfully.
OK
Add New Tourist
Update Tourist Record
Delete Tourist Record
Show Tourist Record
Load SQL Script
Type here to search

10.



VIDEO DEMONSTRATION:

.mp4

bandicam 2025-01-19 06-00-36-282.mp4

Conclusion:

The Pakistan Tourism Management System is a robust, user-friendly DBMS that meets the specified requirements. It showcases the integration of database design principles, normalization, and functional CRUD operations in a real-world application. The portal enhances usability and ensures efficient data management.

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